

Implementing TDMA Slot Allocation in GSM

1. Aim

To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

2. Software Used

MATLAB

3. Code

```
clc;
clear;
close all;

% Number of Users
users = 4;

% Number of Time Slots
slots = 8;

% Slot Allocation
allocation = mod(0:slots-1, users) + 1;

% Display Slot Allocation
disp('TDMA Slot Allocation');
disp('-----');

for i = 1:slots
    fprintf('Time Slot %d --> User %d\n', i, allocation(i));
end

% Plot 1: Bar Graph
figure;
subplot(2,1,1)
bar(1:slots, allocation)
grid on
title('TDMA Slot Allocation')
```

```
% Number of Time Slots
```

```
slots = 8;
```

```
% Slot Allocation
```

```
allocation = mod(0:slots-1, users) + 1;
```

```
% Display Slot Allocation
```

```
disp('TDMA Slot Allocation');
```

```
disp('-----');
```

```
for i = 1:slots
```

```
    fprintf('Time Slot %d --> User %d\n', i, allocation(i));
```

```
end
```

```
% Plot 1: Bar Graph
```

```
figure;
```

```
subplot(2,1,1)
```

```
bar(1:slots, allocation)
```

```
grid on
```

```
xlabel('Time Slot')
```

```
ylabel('User')
```

```
title('TDMA Slot Allocation in GSM')
```

```
% Plot 2: Stem Plot
```

```
subplot(2,1,2)
```

```
stem(1:slots, allocation,'filled','LineWidth',2)
```

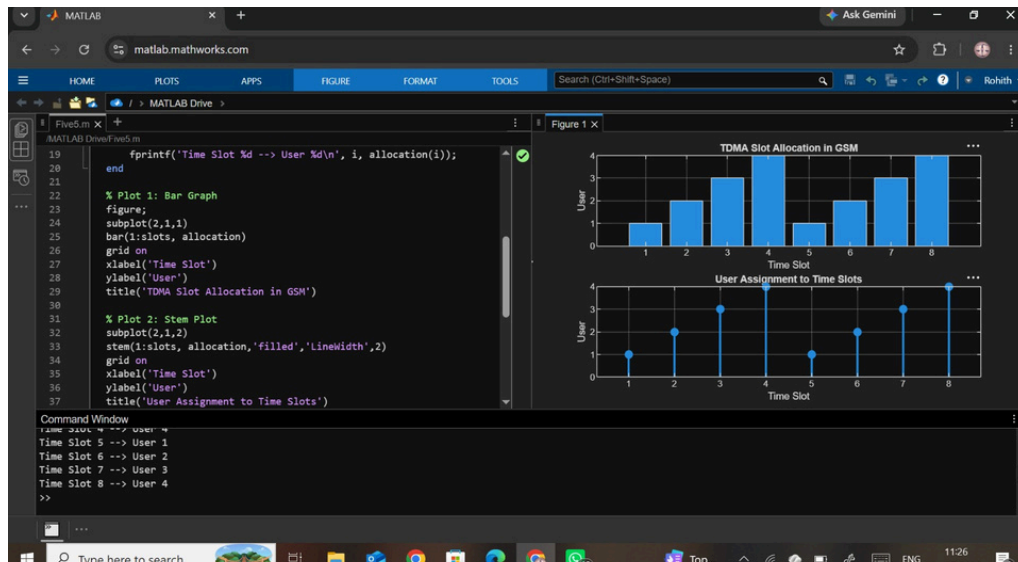
```
grid on
```

```
xlabel('Time Slot')
```

```
ylabel('User')
```

```
title('User Assignment to Time Slots')
```

6. Output Graphs



7. Result

The TDMA slot allocation in a GSM system was successfully implemented using MATLAB. Eight time slots were allocated to four users in a round-robin manner. The generated graphs clearly show the assignment of time slots to users, demonstrating efficient time-sharing and collision-free communication in a GSM TDMA system.
